

PNMF — Basic Architecture Overview

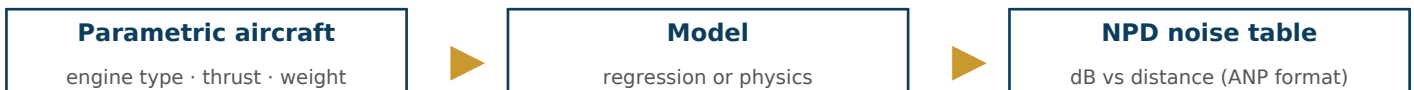
How the Parametric Noise Modeling Framework predicts the noise of a future aircraft · TU Darmstadt / FSR Advanced Design Project

1 · What the framework does

Given a simple description of an aircraft (engine type, number of engines, thrust, weight), PNMF predicts how loud it will be — as a noise-vs-distance table in the exact EASA ANP format — even for concept aircraft that were never built or flight-tested.

Certification noise data only exists for aircraft that already fly. PNMF learns the pattern “**aircraft parameters** → **measured noise**” from the existing fleet, then applies that pattern to a new design so it can be assessed inside the standard ANP / Doc-29 noise pipeline.

2 · What goes in, what comes out



Input — the parametric aircraft (a handful of numbers, no test flight needed): engine type (jet / turboprop / piston), number of engines, thrust per engine, MTOW, MLW, and ICAO noise chapter.

Output — an NPD table in EASA ANP layout: the noise level (dB) at **10 fixed slant distances** (200 ft ... 25 000 ft), for several engine power settings, produced for each noise **metric** (SEL, L_{Amax}, EPNL, PNLT_M) and each **operating mode** (Departure, Approach).

3 · The data — what we train and test on

Database used	EASA ANP v2.3 (the “LEGACY” CSV export)
What it contains	~111 real jet aircraft, each with measured noise curves + its engine and weight parameters (2 776 measured data points total)
How it is loaded	ANPDatabase (pnmf/anp.py) reads one sqlite datastore built from the v2.3 files
Role	the ONLY data the models learn from AND are tested against

Engine type: the trained and validated fleet is **jet turbofans** (the 111 jets). Engine type is still an input field so a turboprop or piston concept can be flagged, but the accuracy numbers in §5 are for jets.

About ANP v6.3: the newer *EASA_ANP_database_v6.3* is present in the repo (03_data/EASA_ANP_database_v6.3/) but is **not used anywhere** — not for training, not for evaluation, not for comparison. It is a newer, xlsx-only export whose schema the current loader cannot read; adding a converter is listed as **future work**. Everything here runs on **v2.3**.

4 · How the prediction is made — two independent models

Route A — Regression model (the default): RandomForest. We fit a RandomForest regression that learns, from the 111 real aircraft, the mapping *aircraft parameters* → *noise levels* (one model per metric and mode). To predict a new aircraft we feed its parameters in and read the curve out. A RandomForest's prediction is the **average of many decision trees**, and the spread between them gives a built-in uncertainty:

$$L_{\text{pred}} = (1 / N_{\text{trees}}) \cdot \sum_k L_k \quad \sigma = \text{std}_k(L_k)$$

A physical rule is then enforced: noise must **not increase with distance**.

Route B — Physics model (independent cross-check). Instead of learning, this route **computes** the noise from first principles: the jet, the fan, and the airframe each generate sound through known acoustic formulas, which are summed and propagated to the listener with distance and air absorption:

$$\text{SPL}(\text{at distance } r) = \text{SPL}(\text{at } 1 \text{ m}) - 20 \cdot \log_{10} r - (\text{air absorption}) \cdot r$$

It is tuned **once** on the A320 and then **frozen**, so every other prediction is a genuine test. (It produces SEL and L_{Amax}; EPNL/PNLT_M are out of scope for this route.)

How we achieved it — the build pipeline: (1) load the real ANP v2.3 fleet → (2) turn each aircraft into its parameter vector → (3) fit the RandomForest per metric/mode → (4) validate leave-one-aircraft-out → (5) cross-check against the physics route → (6) gate every prediction with automatic quality checks before storing.

5 · How we know it works

We use **leave-one-aircraft-out** testing: hide one aircraft, train on the other 110, predict the hidden one **from its parameters alone**, and measure the error. No model ever sees the aircraft it is graded on.

Evidence	Result	Meaning
Leave-one-out error, regression (n = 111)	~4-5 dB avg, 2-3 dB median	reliable within a few dB across the whole fleet
Best vs simplest model (bake-off)	RandomForest 2.91 dB wins	why RandomForest is the default
Physics route vs reality (out-of-sample)	median 2.8 dB	matches the 3-4 dB “good agreement” band in the literature
Two routes on an unseen concept	agree to ≈ 0.2 dB	independent methods converge — strong confidence
Blind test on a real Boeing 737-800	1-1.5 dB approach, 3-4 dB departure	end-to-end accuracy on a fully held-out aircraft

Bottom line. Trained only on today's jet fleet, PNMF predicts a withheld aircraft's noise table to within a few dB, and a completely separate physics model confirms it — evidence that it generalises to new aircraft that have no certification data, which is exactly what it was built for.